JURONG JUNIOR COLLEGE
JC2 Preliminary Examination 2018

CANDIDATE NAME	**Teacher's copy**		
CLASS		INDEX NUMBER	

BIOLOGY — 8876/02

Paper 2 Structured Questions and Free-response Questions

23 August 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Section B	
Total	

This document consists of **24** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section

1 Fig. 1.1 shows an electron micrograph of mitochondria cross-sections.

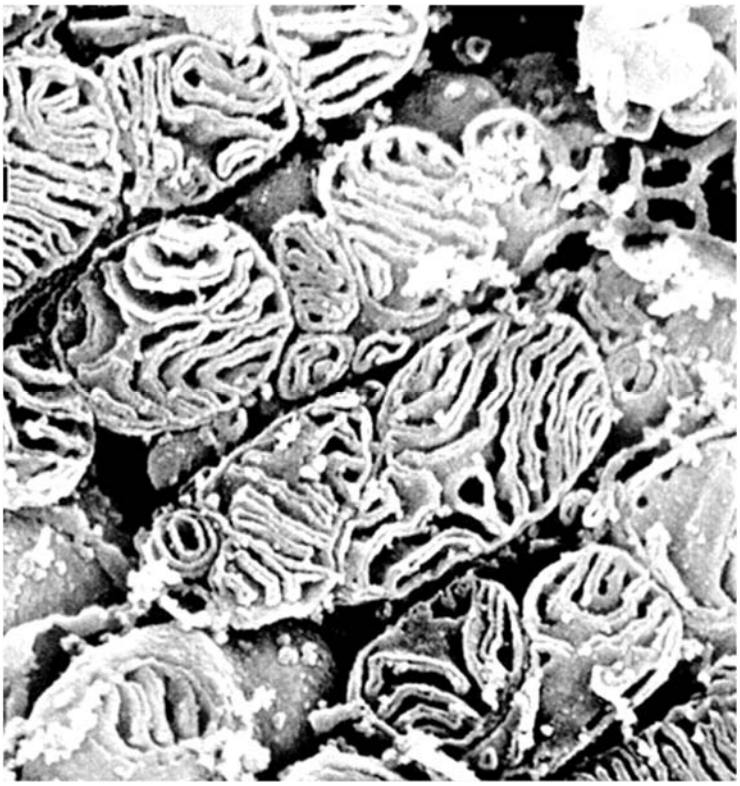

Fig. 1.1

(a) With reference to Fig. 1.1, state one visible feature of the mitochondrion and explain how this feature is adapted for the mitochondrion's function. [3]

Visible feature:

1. **The mitochondrion has a <u>highly folded inner mitochondrial membrane</u> / has <u>numerous infoldings/cristae</u> ;**

Adaptation:

2. **which provides a (large) surface (area) for the <u>attachment of enzymes / electron transport chains / electron carriers / ATP synthase</u> ; [Reject: for H^+ diffusion]**
3. **involved in oxidative phosphorylation <u>for ATP synthesis</u>. ;**

OR

Visible feature:

4. **The mitochondrion has a <u>narrow intermembrane space</u> / <u>Compartmentalisation</u> of the mitochondrion <u>by the inner mitochondrial membrane</u> into regions like the intermembrane space ;**

Adaptation:

5. **facilitates the <u>accumulation / concentration of H^+</u> leading to the (rapid) establishing of a (steep) electrochemical / proton gradient required to ;**
6. **drive <u>ATP synthesis</u>. ;**

[must make correct match between Visible Feature and Adaptation]

Fig. 1.2 shows the molecular structure of carbonyl cyanide-4-(trifluoromethoxy) phenylhydrazone (FCCP). FCCP is a respiratory poison that binds protons and transports them across the phospholipid bilayer of the inner mitochondrial membrane. Protons alone are unable to diffuse freely in this manner.

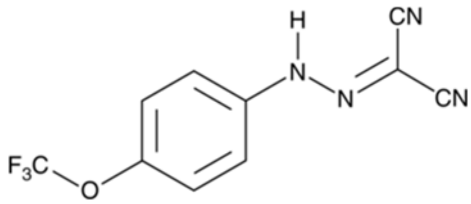

Fig. 1.2

(b) Explain, in relation to their properties, why FCCP readily diffuses across the phospholipid bilayer while protons do not. [2]

1. **FCCP is a small, hydrophobic non-polar molecule and readily diffuses across the hydrophobic core of the phospholipids bilayer. FCCP is hence soluble in lipids. (all subjects required) ;**
2. **Protons are charged and are repelled by the hydrophobic core of the phospholipid bilayer of the inner mitochondrial membrane. Thus, protons cannot diffuse readily. ;**
   [Mark once for *"hydrophobic core"*]

In the presence of FCCP, the rate of ATP synthesis during respiration is significantly diminished. The rate is further reduced if oxygen becomes unavailable. In such conditions where oxygen is depleted, cells continue to oxidise glucose to form pyruvate and sustain ATP synthesis.

(c) (i) Identify the key stages where ATP synthesis occurs via substrate level phosphorylation. [1]

glycolysis and Krebs cycle ;

(ii) State the location where the biochemical pathway enabling the continual oxidation of glucose to form pyruvate occurs. [1]

Cytosol / cytoplasm ;

(iii) Explain how ATP synthesis can be sustained in the absence of oxygen in yeast cells. [3]

1. **ATP in yeast can be sustained through anaerobic respiration, where glycolysis occurs followed by alcoholic fermentation. ;**
2. **Regeneration of NAD (during alcoholic fermentation) allows glycolysis to continue ;**
3. **resulting in a net gain of 2 ATP per molecule of glucose via substrate level phosphorylation. ;**

[Total: 10]

2 Fig. 2.1 shows a representation of a starch molecule. Starch is made up of amylose and amylopectin.

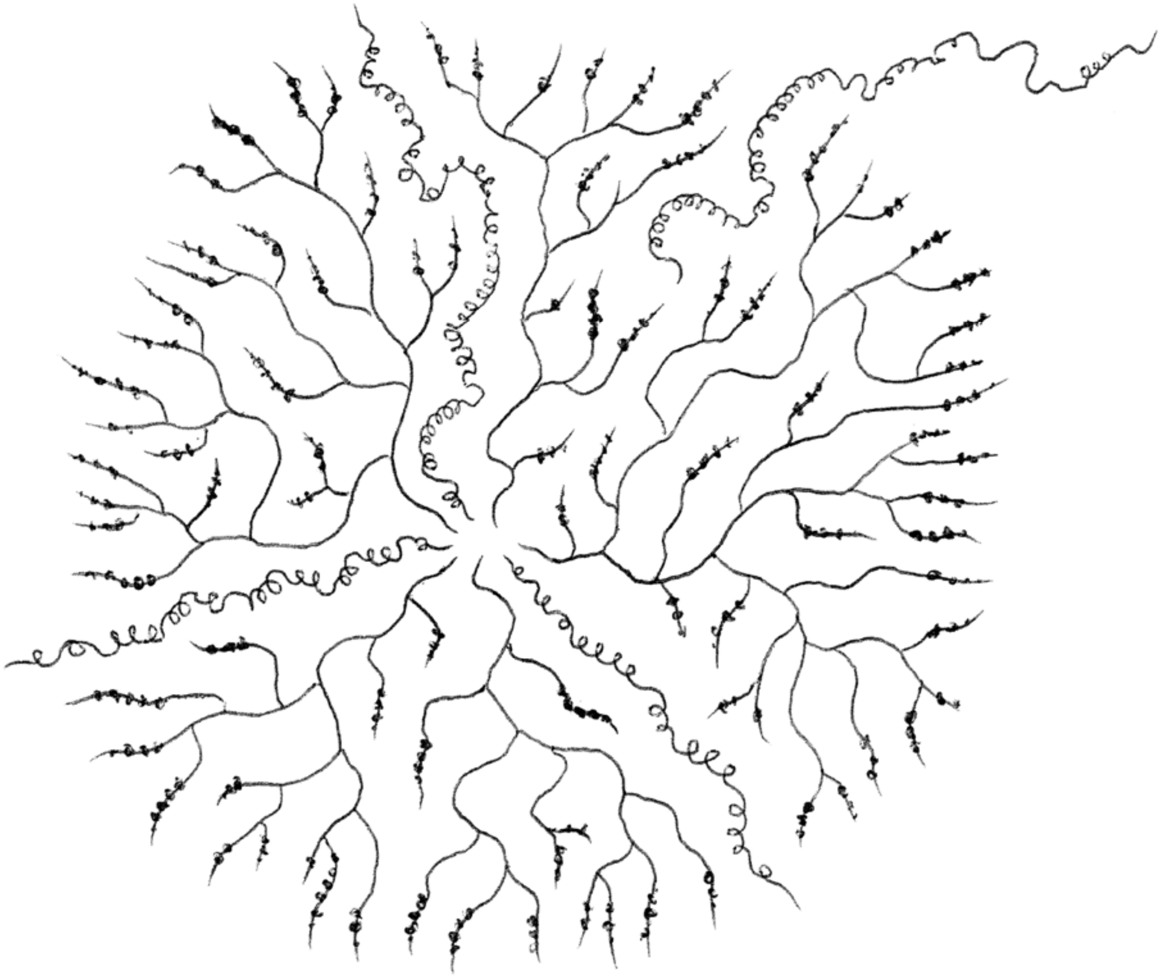

Fig. 2.1

(a) On Fig. 2.1, label amylose and amylopectin. [2]

(b) Explain how the structure of amylopectin is related to its role in living organisms. [2]

1. **Amylopectin is highly branched / many side chains formed by α-1, 6 glycosidic bonds, hence there are more branched ends / sites for hydrolytic enzymes to act on at any one time / can be hydrolysed by hydrolytic enzymes more efficiently; hence amylopectin acts as an accessible source of glucose**
2. **The glycosidic bonds (between monomers) in amylopectin can be hydrolysed easily, resulting in the release of a large number of glucose monomers; hence amylopectin acts as an accessible source of glucose**
3. **Amylopectin is a large (polysaccharide) / long chain (of glucose residues), which makes it insoluble in water; making amylopectin a suitable energy storage molecule within the cell**
4. **Insoluble nature of amylopectin prevents its diffusion out of the cell, hence it does not affect/AW the water potential / has no osmotic effect (within the cell); making amylopectin a suitable energy storage molecule within the cell**
5. **Due to the compact structure / packing of many glucose molecules per unit volume, amylopectin serves as a good storage molecule (TEACH) / good energy source in animals; as a lot of energy in the form of ATP is released when amylopectin is hydrolysed**

(c) Describe one structural difference between cellulose and amylopectin. [1]

1. **a polymer of β-glucose vs a polymer of α-glucose;**
2. **successive β-glucose residues are rotated 180° with respect to its adjacent residue vs successive α-glucose residues are not rotated 180° with respect to its adjacent residue;**
3. **consists of (a long chain of) β-glucose residues joined by β-1,4 glycosidic bonds vs (a backbone of) α-glucose residues held together by α-1,4 glycosidic bonds, and is highly branched/with side chains formed by α-1,6 glycosidic bonds;**
4. **unbranched/linear vs branched;**
5. **hydrogen bonds are formed between the –OH groups of neighbouring cellulose chains lying in parallel (resulting in cross-linking) vs most of the –OH groups are projected into the interior of the helix, hence there are no free –OH groups to form hydrogen bonds with between neighbouring amylopectin;**
6. **hydrogen bonds / cross-linking are formed between neighbouring cellulose chains (lying in parallel) vs hydrogen bonds are not form / no cross linking between neighbouring amylopectin;**
7. **In cellulose, –OH groups projecting outwards from each cellulose chain in all directions vs in amylopectin, most of the –OH groups are projected into the interior of the helix;**

[Total: 5]

3 Fig. 3.1 shows the origin and development of a B cell (a type of white blood cell) from a stem cell.

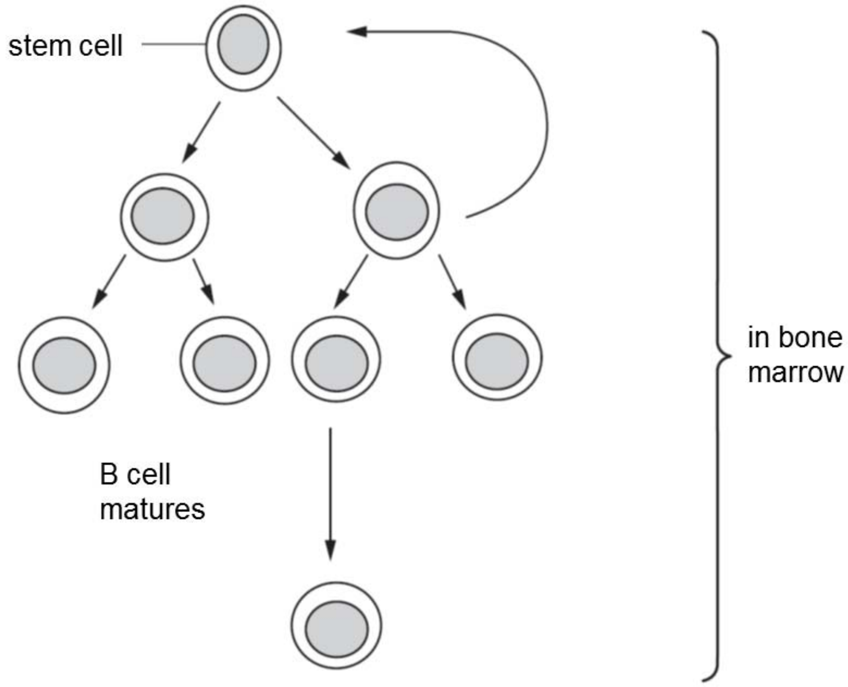

Fig. 3.1

(a) Explain what is meant by a *stem cell*. [2]

1. **Stem cells are capable of <u>remaining undifferentiated</u> (and unspecialised) for a long time ;**
2. **Stem cells are capable of <u>dividing indefinitely</u>/<u>renewing itself many times</u> by <u>mitosis</u> ;**
3. **Stem cells can <u>undergo differentiation</u>, giving rise to <u>specialised cell types</u> (upon receiving appropriate molecular signals) ;**

During B cell development, specific cell surface proteins known as CD proteins are produced and incorporated into the cell surface membrane.

Fig. 3.2 shows the stages involved in the synthesis of a CD protein in a B cell.

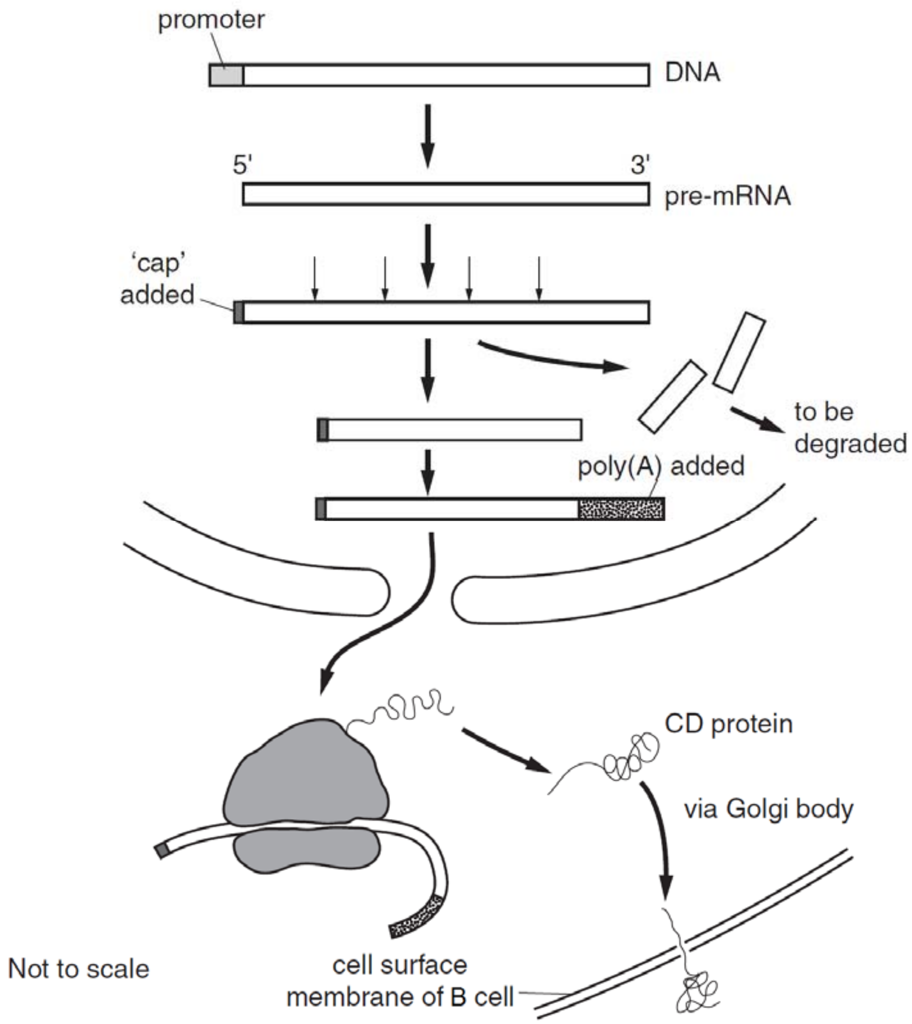

Fig. 3.2

(b) With reference to Fig. 3.2, outline the process:

(i) occurring in the nucleus of the B cell to produce pre-mRNA [3]

1. **<u>Transcription</u> is the synthesis of an RNA molecule with a base sequence complementary to a section of DNA in the nucleus ;**
2. **<u>RNA polymerase</u> recognises and binds to the (core) promoter, forming a transcription initiation complex together with general transcription factors ;**
3. **<u>Free RNA nucleotides</u> align opposite the template strand via <u>complementary base-pairing</u> ;**
4. **(base-pairing) Cytosine pairs with guanine, guanine with cytosine, thymine with adenine and adenine with uracil, forming hydrogen bonds with complementary DNA bases ; (any 2)**
5. **RNA polymerase joins these RNA nucleotides together by catalysing the formation of a <u>phosphodiester bond between RNA nucleotides</u> ;**

(ii) in forming the translation initiation complex. [3]

1. **A <u>small ribosomal subunit</u> recognises and <u>binds to the 5' end of the mRNA</u> and travels along the mRNA until it <u>reaches the first AUG</u> / <u>start codon</u>;**
2. **A special <u>initiator tRNA</u> carrying the <u>amino acid methionine (Met)</u> / <u>anticodon UAC</u>, binds to the start codon AUG on the mRNA;**
3. **The union of mRNA, initiator tRNA, and a small ribosomal subunit is followed by the attachment of a <u>large ribosomal subunit</u>, completing a <u>translation initiation complex</u>;**
4. **Proteins called <u>initiation factors</u> and <u>GTP</u> are required to bring all these components together;**

(c) Burkitt's lymphoma is a cancer of B cells. It is caused by a mutation that usually involves chromosome 8.

(i) Name one causative agent of mutation in chromosome 8 of Burkitt's lymphoma. [1]

5. **(Exposure to) ionising radiation, e.g. X-rays / gamma rays / UV radiation /** nuclear radiation;
6. **(Exposure to) chemical carcinogens, e.g. tobacco/tar in cigarettes;**
7. **Infection by viruses, e.g. human papilloma virus (HPV);**

(ii) Describe the key changes of two types of genes and their effects on cancer development. [2]

1. **Gain-of-function mutation in (one copy of) proto-oncogenes results in <u>overstimulation of cell cycle/uncontrolled cell division</u> ;**
2. **<u>Loss-of function mutation in (both copies of) tumour suppressor genes</u> results in <u>inappropriate cell cycle progression/cell cycle not being arrested</u> ;**
3. **Mutation in genes which control regulatory checkpoints of the cell cycle results in inappropriate cell cycle progression ;**
4. **Mutation in the genes involved in apoptosis causes cells to evade programmed cell death (even when the DNA of the cells are irreparably damaged) ;**
5. **Up-regulation/Activation of genes coding for telomerase prevents the progressive shortening of telomeres with each round of DNA replication, such that cancer cells (have limitless replicative potential and) can divide indefinitely ;**

[Total: 11]

4 Galactosaemia is a rare genetic disease in which the build-up of the monosaccharide galactose can result in an enlarged liver, kidney failure and brain damage. Galactose is produced in the body from the digestion of the sugar lactose, found in milk.

Galactosaemia is caused by a recessive mutation of the *GALT* gene. The normal dominant allele codes for an enzyme which converts galactose to glucose.

(a) **(i)** Suggest how a person with galactosaemia can minimise damage to the liver, kidney and brain. [1]

- **consume, less / no, milk / lactose / (named) dairy products;**

(ii) Explain how a mutation in the *GALT* gene could result in a change in the enzyme responsible for the metabolism of galactose. [4]

1. **Single base pair substitution has occurred in the DNA / gene;**
2. **Different codon is coded for in the mRNA, resulting in a missense mutation;**
3. **(resulting in a change in amino acids with different properties due to different R groups and) a change in polypeptide sequence/ primary structure;**
4. **change in specific 3D conformation of protein;**
5. **change in allosteric site / active site;**
6. **resulting in a non-functional enzyme / does not convert galactose (to glucose);**

(b) If the phenotypes of parents are known, the probabilities of having a child with galactosaemia, an unaffected child (healthy, not a carrier) or a child who is a carrier can be calculated.

Complete Table 4.1 to show the results of these calculations. [2]

Table 4.1

parent 1	parent 2	percentage probability of having a child with galactosaemia	percentage probability of having an unaffected child	percentage probability of having a child who is a carrier
unaffected	carrier	0	50	50
carrier	carrier	....	....	....
unaffected	has galactosaemia	....	....	....
carrier	has galactosaemia	50	0	50

carrier	carrier	25	25	50	;
unaffected	has galactosaemia	0	0	100	;

[Total: 7]

5 Silvereyes (*Zosterops lateralis*) are a species of Australian songbirds that can be found in both rural and urban environments. These birds produce contact calls to inform one another of danger or newly discovered food sources. The pitches and speed (measured in number of syllables per second) of Silvereyes' contact calls were identified to be genetically associated.

Studies on Silvereyes living in increasingly denser Australian cities have revealed that the birds made contact calls with much higher mean pitches and slower mean speeds with longer pauses between calls over the years.

When measured in the past, the mean pitch of the Silvereye call overlapped significantly with the pitches of road and air traffic making the calls indistinguishable from urban noise. Sound-reflecting buildings also diminished the clarity of the bird calls.

Fig. 5.1 shows a Silvereye.

Fig. 5.1

(a) Explain how natural selection may have led to a change in the mean pitch and speed of the Silvereye calls in the city. [5]

1. **<u>Spontaneous mutation results in genetic variation</u> (in alleles coding for Silvereye contact call pitch and speed) within the population of Silvereyes living in the city ;**
2. **Birds that do not make clearer calls hence <u>die from predation / lack of food</u>. ;**
   - **Selection Pressure: Urban noises such as road and air traffic and sound-reflective buildings muffled the Silvereye contact call, diminishing the success of birds communicating danger and newly discovered food sources among themselves.**
3. **On the other hand, Silvereyes that made calls at a <u>higher pitch and slower speed</u> are at <u>selective advantage</u>. ;**
4. **Silvereyes selected for these advantageous traits <u>survive to maturity</u>, <u>mate/reproduce</u> and <u>passed down favourable alleles (that code for higher-pitched and slower contact calls) to their offspring</u> / ORA. ;**
5. **Over time, there is a <u>change in the allele frequencies in the population</u> for contact call pitch and speed, leading to evolution of a higher mean pitch and slower speed with pauses in Silvereye calls. ;**

(b) An analysis of ice cores from the Arctic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 5.1 shows the concentrations of carbon dioxide measured in ice cores, dated between 1000 and 2000 AD.

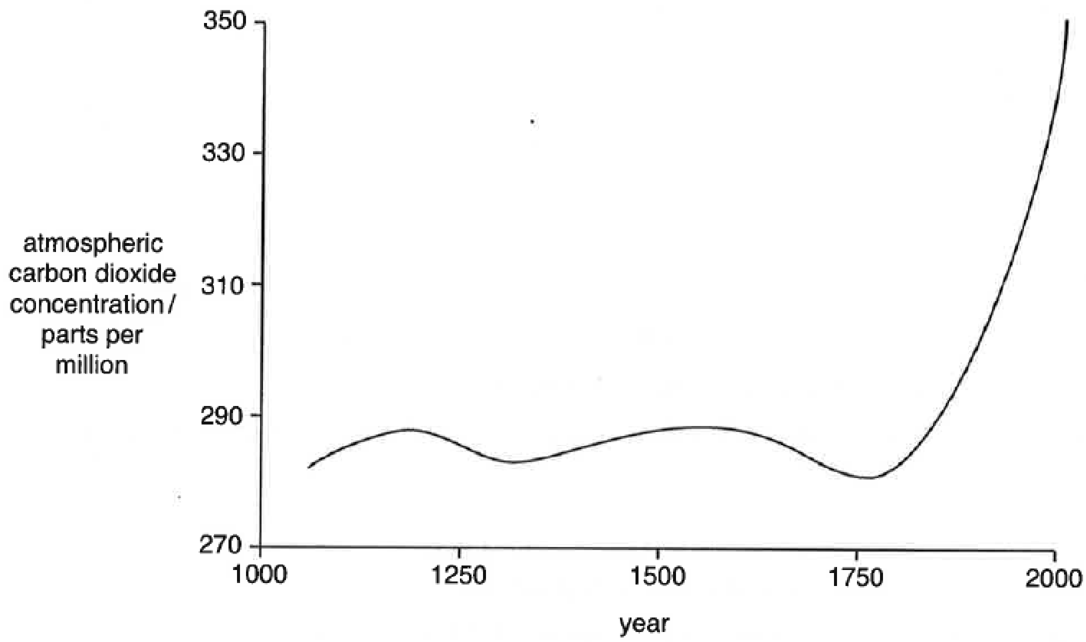

Fig. 5.1

(i) Describe the trend in Fig. 5.1. [2]

1. **The atmospheric carbon dioxide concentration stayed relatively constant at around 285 parts per million in year 1060 and increased exponentially/sharply in year 1750 to reach 350 parts per million in year 2000;**

   1m for constant, increased sharply
   1m for pair of figures

(ii) Atmospheric carbon dioxide concentrations show regular annual variations. Suggest one reason for this. [1]

1. **There are (four) seasons/seasonal changes in a year / atmospheric carbon dioxide concentrations is lower in summer and higher in winter;**

(c) Fig. 5.2 shows that, over the same period of time, the average surface temperature of the Earth has shown a similar pattern of change. The increasing concentrations of carbon dioxide is thought to be responsible for the increase in temperature over the last 100 years. This is referred to as the enhanced greenhouse effect.

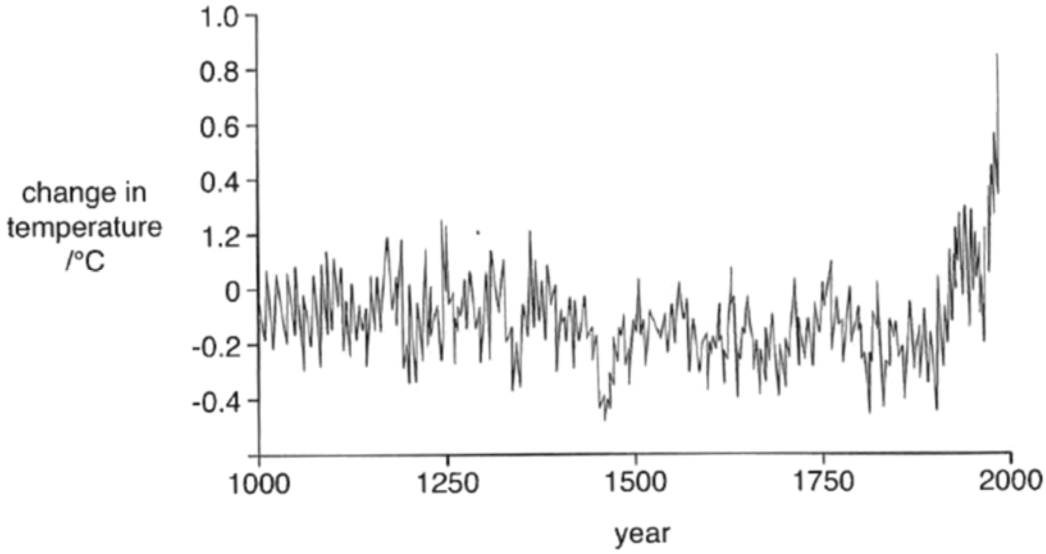

Fig. 5.2

(i) Describe one way in which the data in Fig. 5.2 resembles the data in Fig. 5.1 and one way in which it is different. [2]

1. **Similarity: The change in temperature stayed relatively constant before increasing exponentially/sharply**

2. **Difference: there are more variations/fluctuations in the changes in temperature and less variations/fluctuations in atmospheric carbon dioxide concentrations;**

(ii) Explain the human activities that have contributed to the enhanced greenhouse effect. [2]

1. **Burning of fossil fuels due to increasing energy usage releases large amounts of stored carbon into the atmosphere as carbon dioxide (CO_2) and is the major source of CO_2 emission;**
2. **Deforestation causes a net reduction in carbon storage as forested areas act as carbon sinks and results in CO_2 emission when forests are burnt, increasing the level of CO_2 in the atmosphere;**

[Total: 12]

Section B

Answer **one** question in this section

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)** and **(b)**, as indicated in the question.

6 **(a)** The organelles of the endomembrane system in animal cells are related through direct contact or by the transfer of membrane segments as vesicles.

Outline the functions of the organelles of the endomembrane system and state the structural similarities between these organelles. [9]

(Max 7 from pt 1 – 14)

Functions of the rough ER

1. **sites of protein synthesis;**
2. **biochemical modification takes place in cisternal space / proteins may be modified by enzymes in the cisternal space of ER that add carbohydrate chains (or lipids) to them – glycosylation;**
3. **serves as the intracellular transport system which transports the synthesised/modified proteins to other compartments within the cell by transport vesicles budding off from the ER membrane;**
4. **The membrane of transport vesicles (formed from rough ER membrane) also replenishes the membrane of Golgi body.;**

Functions of the smooth ER

5. **responsible for the synthesis of lipids (e.g. phospholipids and steroid hormones);**
6. **involved in carbohydrate metabolism in the liver (glycogenolysis);**
7. **detoxification of drugs and other toxic substances in the liver;**
8. **storage of calcium ions necessary for muscle contraction;**

Function of Golgi body

9. **Further modify, sort and package proteins / products of ER into vesicles before transporting to other parts of the cell and/or out of the cell / targeted for other destinations;;**
10. **Formation of secretory vesicles containing matured proteins which move to the cell surface and fuse with the plasma membrane, releasing their contents out of the cell via the process of exocytosis;;**
11. **Formation of lysosomes containing hydrolytic enzymes (e.g. proteases, nucleases, lipases);;**

Function of Lysosomes

12. **Intracellular digestion – to digest material which the cell consumes from the environment through phagocytosis;**
13. **Autophagy – to digest parts of the cell such as damaged or worn-out organelles;**
14. **Autolysis – self-destruction of a cell by releasing the hydrolytic enzymes of all lysosomes within the cell.;**

Structural similarities

15. **Presence of (single) membrane / phospholipids bilayer;**
16. **Presence of proteins/enzymes within organelles;**
17. **Presence of fluid-filled space;**
18. **Association with cytoskeleton;**

QWC:
Good spread of knowledge communicated without ambiguity to include:
At least 2 organelles' functions and at least 1 MP on the structural similarities.

(b) A 75kg human consists of ten times more prokaryotic cells than eukaryotic cells, with a total prokaryote mass of at least 1kg. This assembly of prokaryotic cells is known as the human prokaryotic microbiome community.

Outline the differences between typical prokaryotic and eukaryotic cells and state the methods by which these differences can be shown. [6]

Differences (max 5)

	Features	**Eukaryotes**	**Prokaryotes**
1.	**Size**	**Large, 10-100μm**	**Small/<1 μm**
2.	**Nucleus/nuclear envelope**	**Present**	**Absent**
3.	**Cell wall**	**Absent in animal cell**	**Present in prokaryotic cell**
4.	**Membrane bound organelles**	**Present**	**Absent**
5.	**Ribosomes**	**80S/large/22nm**	**70S/small/18nm**
6.	**DNA conformation**	**Linear**	**Circular**
7.	**Introns**	**Present**	**Absent**
8.	**Origin of replication *(ori)***	**Many**	**Few**
9.	**Compaction of DNA**	**Highly condensed**	**Less condensed**
10.	**Size of genome**	**Large**	**Small**
11.	**Histones**	**Present**	**Absent**
12.	**Component in cell wall**	**Cellulose in plant cell**	**Peptidoglycan in bacterial cell**
13.	**Peptidoglycan**	**Absent**	**Present**

Pt 12 & 13: award once

Methods:

14. Light microscope;
15. Use of stain;
16. Electron microscope;
17. Centrifugation;
18. Chemical analysis;
19. DNA analysis/PCR/DNA sequencing/electrophoresis;
20. Other valid methods;

7 **(a)** Variation exists in individuals of the same species in a population due to a number of different reasons. Describe what causes variation and why it is important in natural selection. [9]

<u>Causes of Variation (Max 7)</u>

1. **Variation is due to mutation, meiosis and sexual reproduction ;**

<u>Gene Mutation</u>

2. **Gene mutations are defined as changes in the sequence of DNA nucleotides in the gene;**
3. **Due to exposure to chemical carcinogens such as tobacco in cigarette smoke / ionising radiation such as UV-rays, X-ray / viruses / errors during DNA replication or repair etc.;**
4. **Base pair substitution / addition / deletion may occur, resulting in nonsense / silent / missense / frameshift mutation; (max 1)**

5. **Base-pair substitution – replacement of one nucleotide base pair with another base-pair in a gene, resulting in missense mutation/nonsense mutation/significant change in the encoded protein OR silent mutation/no/little effect on the encoded protein;**

6. **Base-pair addition – insertion of one or more nucleotide base pairs in a gene, resulting in frameshift mutation, whereby the reading frame is altered / nucleotides downstream from the mutation are improperly grouped into incorrect codons and read in different sets of threes;**
   OR
7. **Base-pair deletion – loss of one or more nucleotide base pairs in a gene, resulting in frameshift mutation, whereby the reading frame is altered / nucleotides downstream from the mutation are improperly grouped into incorrect codons and read in different sets of threes;**

Max 2 for MP 4-7

<u>Chromosomal aberration</u>

8. **Chromosomal aberration can be categorised as numerical aberration (the change in chromosome number) or structural aberration to chromosomes;**

9. **Translocation – a section of chromosome breaks off from one chromosome and becomes attached to another chromosome;**
10. **Duplication – a section of a chromosome replicates such that a set of gene loci is repeated;**
11. **Deletion – a chromosome breaks at two points, the middle portion is displaced with the two ends joining together;**
12. **Inversion – a chromosome breaks at two locations and the middle portion flips through 180° before rejoining;**

13. **Aneuploidy is a condition in the nucleus where there are one or several chromosomes more than or less than the diploid number of chromosomes;**
14. **Aneuploidy can result from non-disjunction during anaphase / when a haploid gamete fuses with a gamete carrying n-2, n-1, n+1 or n+2 chromosomes;**
15. **Polyploidy is a condition of the nucleus where there are three or more times the haploid number of chromosomes, e.g. 3n, 4n and 5n;**
16. **It can result from non-disjunction, the fusion of a diploid gamete with a normal haploid gamete giving a triploid nucleus;**

<u>Meiosis</u>

17. **In meiosis, crossing over between non-sister chromatids of homologous chromosomes occurs during prophase I;**
18. **Independent assortment of homologous chromosomes during metaphase I and chromatids during metaphase II;**
19. **Results in new combination of alleles;**

<u>Sexual reproduction</u>

20. **Random fusion of gametes during sexual reproduction;**
21. **Results in new combination of alleles (award once);**

<u>Importance in Natural Selection</u>

22. **Variation describes the <u>differences in characteristics</u> / means the presence of different characteristics;**
23. **due to presence of different alleles in the different individuals;**
24. **resulting in differential reproductive success / different survival rates;**
25. **Variations in characteristics are subjected to selection pressure from the environment;**
26. **~~There can be continuous / discontinuous variation;~~**
27. **Due to interaction of genotype and environment;**
28. **variants with favourable characteristics will survive to maturity, reproduce and pass down their favourable alleles to their offspring;**
29. **Those with unfavourable characteristics die and fail to do so;**

QWC:
Good spread of knowledge communicated without ambiguity to include:
At least 2 causes of variation and at least 1 MP on the importance.

(b) Discuss, using known examples, how limiting factors can influence the rate of various biological processes. [6]

Definition

1. **A limiting factor is that factor which directly affects/determines the rate of reaction/process if its quantity is changed / when the factor is in lowest/shortest/scarcest supply. ;**

Biological Processes

2. **Biological Process: Enzyme-catalysed Reactions ;**

Factor	How Factor Influences Rate	
3. Substrate concentration	+ illustrate how factor influence rate	;
4. Enzyme concentration		;
5. Temperature		;
6. pH		;

[Max 3]

REJECT: Concentration of competitive or non-competitive inhibitors as limiting factors.

7. **Biological Process: Facilitated Diffusion ;**

Factor	How Factor Influences Rate	
8. (Steepness of) concentration gradient / difference in concentrations of transported substance across a cell / biological membrane	+ illustrate how factor influence rate	;
9. Density of channel/carrier proteins on the membrane		;

10. **Biological Process: Photosynthesis ;**

Factor	How Factor Influences Rate	
11. Light intensity	+ illustrate how factor influence rate	;
12. Temperature		;
13. Carbon dioxide concentration		;

14. **Biological Process: Aerobic Respiration ;**

Factor	How Factor Influences Rate	
15. Glucose / hexose sugars concentration	+ illustrate how factor influence rate	;
16. Oxygen availability		;
17. Temperature		;
18. NAD^+ / FAD^+ / mobile electron carriers availability		;

[Max 3]

19. Biological Process: Population Growth / Bacteria Colony Growth ;

Factor	How Factor Influences Rate	
20. Nutrient availability	**+ illustrate how factor influence rate**	**;**
21. Water availability		**;**
22. Temperature		**;**
23. Competition for resources / Predation / AVP		**;**

[Max 3]

REJECT: Signal Reception / Ligand-Receptor Interactions as a form of biological process that is influenced by limiting factors. Any limits on rates of signal propagation and hence response efficacy is overcome by signal amplification.

QWC:
Clearly communicates limiting factors of 2 different biological processes

[Total: 6]